

Rahul Rawat

rahulrawat2r@gmail.com · 015563603340 · linkedin.com/in/rahulrawat2r · github.com/rahulrawat20022002 · Mannheim, Germany

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Avelios Medical

Hiring Team

Application: Working Student Machine Learning in Munich

Dear Hiring Team,

I am applying for the Working Student Machine Learning position at Avelios Medical in Munich. The brief on supporting the integration of intelligent systems into real world clinical environments, in service of the mission to unlock clinical data for seamless healthcare operations and better patient care, aligns directly with the ML delivery on real healthcare adjacent data I have been shipping over the past year.

My bachelor thesis is a leakage aware machine learning study on a real 768 patient clinical dataset, benchmarking six classifiers with 10 fold cross validation and per model confusion matrices. I caught biologically impossible zero values that the original authors had overlooked, applied IQR based outlier removal and proper imputation, and chose ROC AUC over accuracy on a 65 to 35 imbalanced target to avoid a misleadingly rosy accuracy figure. I wrote it up as an IEEE style paper with an honest limitations section. That is the same clinical data mindset Avelios' modular hospital information system needs.

At eRay GmbH I built an end to end recursive time series pipeline forecasting four water quality indicators on real sensor data, benchmarking six models head to head including Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost, and Prophet, and used CatBoost multi quantile regression for asymmetric 80 percent prediction intervals. In CreditIQ I lifted the Disparate Impact ratio from 0.79 to 0.88, cut the false negative rate from 44 percent to 16.7 percent, and backed the pipeline with unit tests at 100 percent branch coverage plus a full regulatory write up on EU AI Act and GDPR. That combination of honest ML evaluation and compliance ready delivery is the right base for real clinical integration.

I am proficient in Python and scikit learn, comfortable with modern ML frameworks, and I speak English fluently with German at B1 in progress. I hold the NVIDIA Building LLM Applications With Prompt Engineering, AWS Academy Cloud Foundations, and Google Data Analytics Foundations certificates. I would be glad to align the exact scope with the Avelios team in Munich.

Kind regards,

Rahul Rawat